

Technical Report: Constructing and Deconstructing RSA – The Path to Quantum Cryptography

1. The Quantum Threat

In the accompanying Python implementation, the period r of the function $a^r \equiv 1 \pmod{N}$ was found using a brute-force loop. The algorithm repeatedly tested $r = 1, 2, 3, \dots$ until the congruence was satisfied. This sequential search is practical only for very small RSA moduli. For real RSA keys (2048 bits or larger), the search space becomes astronomically large, making classical period finding computationally infeasible. RSA security relies on this classical bottleneck.

Shor's Algorithm removes this bottleneck by exploiting quantum mechanics. A quantum computer first places a register of qubits into **superposition**, allowing it to represent many candidate exponents simultaneously. It then evaluates the modular exponentiation function for all of these inputs within one coherent quantum state. The resulting amplitudes encode the hidden periodic structure of the function.

Next, the algorithm applies the **Quantum Fourier Transform (QFT)**. Unlike a classical Fourier transform operating on stored numbers, the QFT manipulates probability amplitudes. Through **quantum interference**, amplitudes corresponding to values consistent with the true period reinforce each other (constructive interference), while inconsistent values cancel out (destructive interference). After measurement, the observed result reveals information from which the period r can be recovered with high probability. Once r is known, the classical $\gcd(a^{r/2} \pm 1, N)$ computation recovers the prime factors p and q . Thus, Shor's Algorithm replaces an exponential classical search with a polynomial-time quantum procedure.

2. The Post-Quantum Defense

Post-Quantum Cryptography (PQC) is the field of cryptography that designs algorithms believed to remain secure even against large-scale quantum computers. Rather than relying on integer factorization or discrete logarithms, PQC uses mathematical problems for which no efficient quantum algorithms are known.

One of the leading PQC families is **lattice-based cryptography**, particularly schemes based on the **Learning With Errors (LWE)** problem. In LWE, an attacker is given many linear equations that contain deliberately introduced small random errors (noise). Recovering the underlying secret requires separating the hidden signal from this carefully chosen noise, a problem believed to be computationally hard even for quantum computers.

The added noise is fundamental to security. Unlike RSA, whose weakness is that factoring can be reduced to period finding, LWE has no known reduction to period finding or integer factorization. Consequently, Shor's Algorithm does not provide an efficient attack against lattice-based schemes. Current NIST-standardized PQC algorithms, including ML-KEM (derived from CRYSTALS-Kyber) and ML-DSA (derived from CRYSTALS-Dilithium), rely on lattice problems rather than factorization, making them strong candidates for protecting communications in the quantum era.

Conclusion. The RSA exercise demonstrates that classical security depends on the infeasibility of finding the period r . Quantum computing fundamentally changes this assumption through superposition and the Quantum Fourier Transform. Post-Quantum Cryptography addresses this threat by adopting mathematical problems, such as Learning With Errors, that are not vulnerable to Shor's Algorithm and are currently considered resistant to both classical and quantum attacks.